



Unit 4 · Outcome 2 · Energy Sources

The seven renewables, and why they never run out

KK02 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



What makes a source "renewable"?

01 What makes a source "renewable"? The whole study design splits energy into non-renewable (KK01) and renewable (KK02). The line between them is not "does it pollute" or "is it clean" — it is about rate. The one-sentence definition examiners want: A renewable energy source is replenished by natural processes at a rate equal to or faster than the rate at which it is used, so it is not depleted on a human timescale. Non-renewable sources (coal, oil, gas, uranium) formed over millions of years and are consumed far faster than they re-form — so the stocks

The Mirran Plains zone in 3D

02 The Mirran Plains zone in 3D Spin the zone. Each installation taps a different natural flow. Click a source chip (or the model) to inspect how it works. Drag to orbit. Click a source to inspect · drag to orbit Solar Wind Hydro Tidal Geothermal Biomass/fuel ☞ Auto-spin * Day / ☾ Night 3D not available in this browser — the source detail cards below cover every installation. Use the chips to read each one. Pick a source above to see how the Mirran Plains zone harvests it.

The seven sources, one card each

03 The seven sources, one card each For every source, learn the same four things: the natural flow it taps, the energy-conversion chain to electricity, why it's renewable, and its standout advantage / disadvantage. The conversion chains matter — they connect straight to KK05. Memory hook "Sun-Wind-Water-Tide, Heat-and-Harvest" — solar & wind (intermittent), hydro & tidal (water-driven), geothermal (Earth's heat), then the two harvest sources: biomass (solid) and biofuels (liquid). Seven sources, three families.

3D · The intermittency lab

04 3D · The intermittency lab Renewables don't all behave the same across a day. Drag the time-of-day slider and flip the weather — watch which sources surge, fade or stay rock-steady. This is the single most examined idea in the chapter, and it sets up base/peak load (KK09). 06:00 · Clear & breezy * Clear/breezy ▲ Cloudy/still 🌩 Windy storm 3D not available — use the readout panel below; it shows the same live output values for each source. Time of day 06:00

Biomass vs biofuels — the classic mix-up

05 Biomass vs biofuels — the classic mix-up The study design lists these as two separate sources, and examiners love testing whether you can tell them apart. Both come from recently living matter; the difference is form and processing. Biomass SOLID · burned directly Organic matter burned to release stored chemical energy as heat. Wood, crop stubble, bagasse (sugarcane fibre), forestry residue, even municipal organic waste. Burned to raise steam → turbine → generator. chemical (plant matter) → thermal → kinetic (steam/turbine) → electrical Mirran Plains

Intermittent vs dispatchable — the deciding axis

06 Intermittent vs dispatchable — the deciding axis Beyond "renewable or not", the property that drives real planning decisions is whether a source can deliver power on demand . This single table answers most comparison questions in the chapter. Source Natural flow Renewable? On-demand (dispatchable)? Greenhouse at point of use Solar PV Sunlight Yes No — intermittent (day/cloud) ~Zero Wind Moving air Yes No — intermittent (wind speed) ~Zero Hydro-electric Falling/stored water Yes Yes — fast, storable ~Zero* Tidal Ocean tides Yes Predictable but cyclic ~Zero Geothermal Earth's internal heat Yes Yes — steady baseload Low Biomass This season's plant growth Yes Yes — fuel can be stored CO₂ released (≈reabsorbed) Biofuels Crops → liquid fuel Yes Yes — fuel stored CO₂ released (≈reabsorbed)

Source	Natural flow	Renewable?	On-demand (dispatchable)?	Greenhouse at point of use
Solar PV	Sunlight	Yes	No — intermittent (day/cloud)	~Zero
Wind	Moving air	Yes	No — intermittent (wind speed)	~Zero
Hydro-electric	Falling/stored water	Yes	Yes — fast, storable	~Zero*
Tidal	Ocean tides	Yes	Predictable but cyclic	~Zero
Geothermal	Earth's internal heat	Yes	Yes — steady baseload	Low
Biomass	This season's plant growth	Yes	Yes — fuel can be stored	CO ₂ released (≈reabsorbed)
Biofuels	Crops → liquid fuel	Yes	Yes — fuel stored	CO ₂ released (≈reabsorbed)

Command-word decoder

◆ COMMAND WORDS

07 Command-word decoder Match what the verb is asking for. Tap a word to expand.

P·E·L writing trainer

🔗 P·E·L WRITING

09 P·E·L writing trainer The house writing frame: Point → Explain → Link . Draft a response to the prompt below; the trainer lights up as you hit the ideas examiners reward. Prompt: Explain why solar and wind are described as intermittent, and why this makes a dispatchable renewable like hydro-electricity valuable in the same grid. (4 marks) Point State that solar/wind depend on a natural condition that varies, so output is not constant. Explain Link the variation to the source (no sun at night/cloud; no output when wind is too low/high) and contrast wit



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Good vs bad answer, side by side

10 Good vs bad answer, side by side Question: Compare a solar farm and a geothermal plant as sources of electricity for the Mirran Plains zone, referring to renewability and reliability of supply. (4 marks) **X** Weak response — 1/4 Solar is good because it's clean and free from the sun. Geothermal uses heat from the ground which is also good. Both are renewable so they're better than coal. The Mirran Plains zone should use solar because Australia is sunny. Vague, one-sided, no real comparison. "Both renewable" isn't explained, supply reliability is ignored,



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

Practice exam questions

★ PRACTICE & SELF-MARK

11 Practice exam questions VCAA-style questions totalling 20 marks . Write your answer, reveal the model and marking guide, then self-mark with the stars. Your progress saves automatically. 0% 0 / 20 marks Reveal a model answer, then rate how you went.

One-screen cheat sheet

♦ RECAP / CHEAT SHEET

12 One-screen cheat sheet Renewable, defined Replenished by natural processes at a rate $\geq$ the rate of use, so not depleted on a human timescale. About rate , not cleanliness. The seven Solar, wind, hydro-electric, tidal, geothermal, biomass, biofuels. Three families: sun/wind, water (hydro/tidal), then Earth-heat + harvest. Intermittent Solar & wind — generate when nature allows, not on demand. Output varies with sun/cloud and wind speed. Dispatchable Hydro, geothermal, biomass, stored biofuels — power on demand. Tidal is predictable but cyclic. Biomass

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Solar is good because it's clean and free from the sun. Geothermal uses heat from the ground which is also good. Both are renewable so they're better than coal. The Mirran Plains zone should use solar because Australia is sunny. Vague, one-sided, no real comparison. "Both renewable" isn't explained, supply reliability is ignored, and the geothermal point is dropped. "Compare" needs both sources weighed against the same criteria.

✓ STRONG / MODEL ANSWER

Both are renewable : sunlight and Earth's internal heat are replenished far faster than they are used. They differ on reliability of supply . Solar PV is intermittent — it generates only in daylight and drops under cloud, so its output varies and cannot be guaranteed at peak evening demand. Geothermal taps a continuous heat flow, so it runs steadily day and night as dispatchable baseload . For Mirran Plains, solar offers cheap bulk daytime energy, while geothermal provides the reliable around-the-clock supply solar lacks — so the two are complementary rather than competing. Both sources weighed on the same two criteria, the intermittent/dispatchable contrast is explicit, and it lands a judge